

NNSS Presentation

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Talking Points

- Background and motivation
- Stochastic ordinary differential equations.
 - Finance
- Stochastic partial differential equations.
 - Research

The Questions

- We want to model the evolution of heat of a piece of initially heated wire. How can we account for random outside sources of heat that affect the wires temperature? (SHE)
- We want to model the growth of a cluster of cells undergoing mitosis. How can we include the randomness in our model? (KPZ- equation)
- We are investors looking to make estimations on the return rate of our current investments. Can we make legitimate assumptions on the "randomness" of our returns? (Ornstein-Uhlenbeck process)

Noise

- Essentially, the randomness is accounted for by a random noise, usually Gaussian.
- The noise can be thought of as the derivative of a non-differentiable function. In the the case of a SODE, the derivative of Brownian motion and in the case of a SPDE, the derivative of a Brownian sheet.
- There are more general noises, but it is still helpful to keep the above "rough" idea in our heads.

Stochastic Processes

Definition (Stochastic Process)

A stochastic process is a random variable that is indexed by set, T , which usually represents time:

$$X_t(\omega), \quad t \in T, \omega \in \Omega.$$

One can think a stochastic processes as a two parameter function

$$X_t(\omega) = X(t, \omega) : T \times \Omega \rightarrow \mathbb{R}.$$

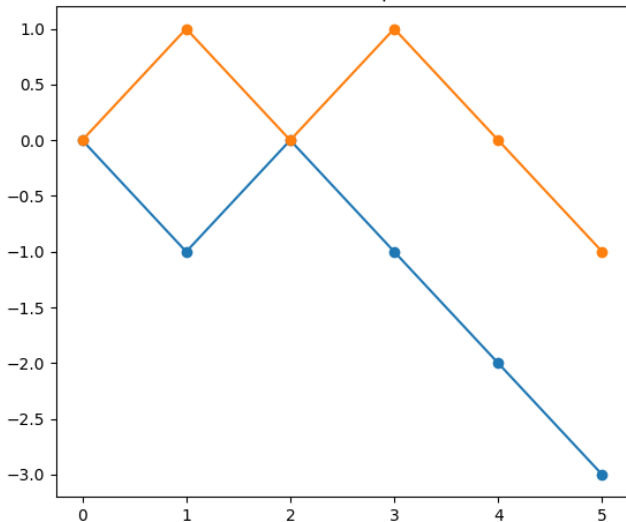
Definition (Sample Path)

The variable ω represents the idea of a single possible outcome. The following function is then referred to as a *sample path*:

$$t \mapsto X_t(\omega).$$

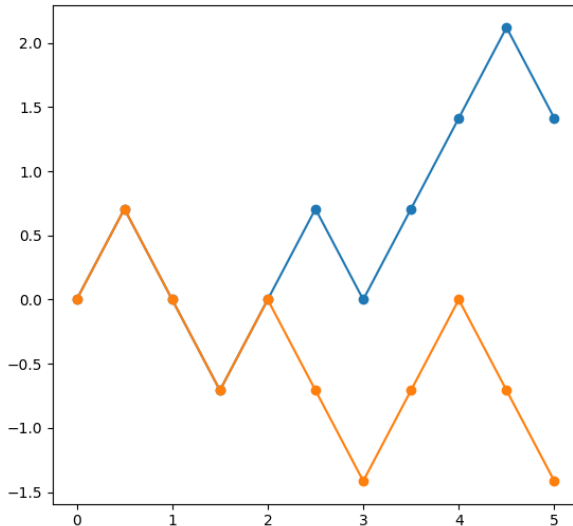
Random Walk Example

2 sample paths of a random walk
jump size = 1.0 and step size = 1.0
number of steps = 5



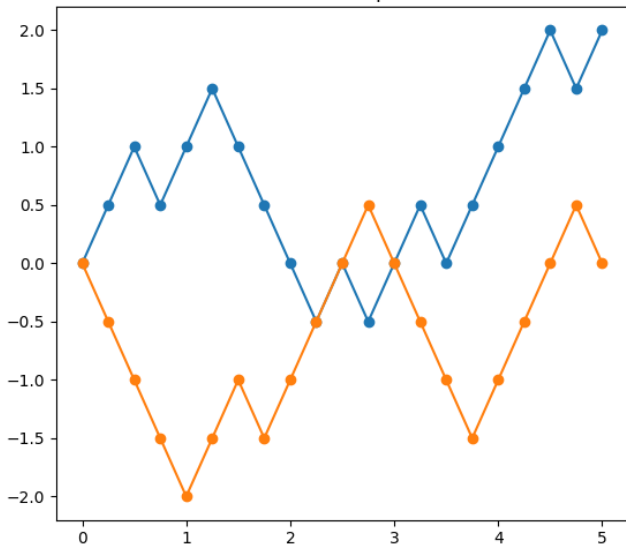
Random Walk Example

2 sample paths of a random walk
jump size = 0.707 and step size = 0.5
number of steps = 10

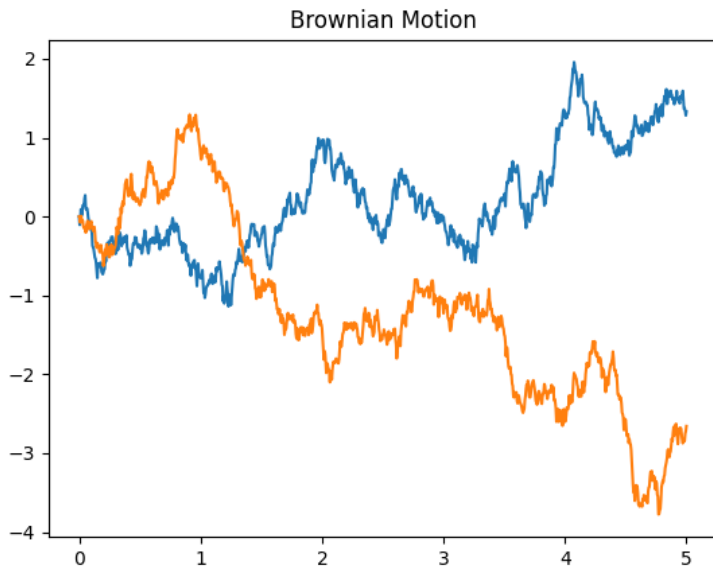


Random walk Example

2 sample paths of a random walk
jump size = 0.5 and step size = 0.25
number of steps = 20

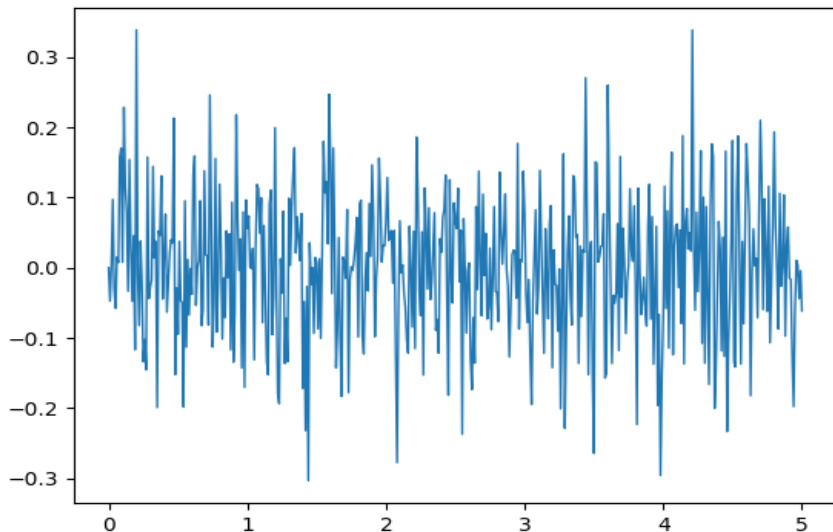


Random Walk Example



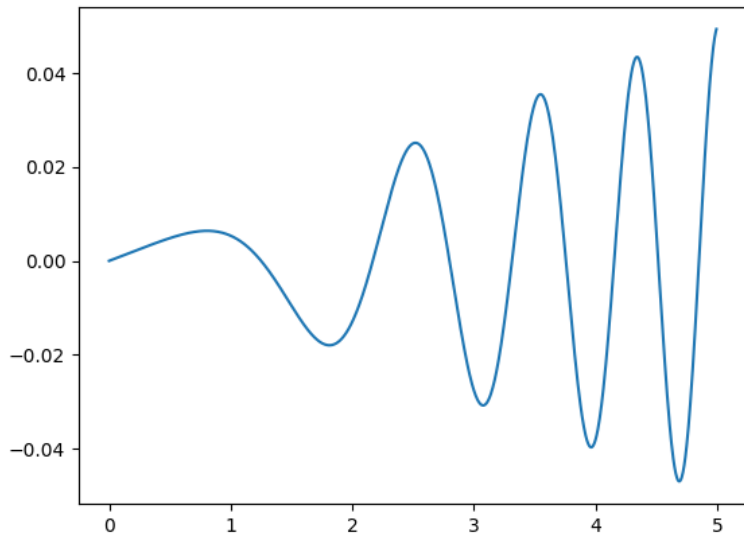
White Noise

White Noise differential: $dB_t \approx B_t - B_{t-\Delta t}$:



Increments of a Smooth Function for Reference

Increments, $df(x_i) = f(x_i) - f(x_{i-1})$, for $f(x) = \sin(x^2)$



What Can We Do with Brownian Motion?

- S_0 = stock price at entry S_t = stock price after t time.

$$S_t = S_0(1 + R_t), \quad R_t := \frac{S_t - S_0}{S_0} \text{ is the return rate over the period } [0, t].$$

- We want to make predictions on the return rate.

$$R_t \sim \text{Normal}(\mu t, \sigma^2 t), \quad \mu \in \mathbb{R}, \sigma \in (0, \infty).$$

- We can model this with a stochastic differential equation.

$$R_t = \frac{dS_t}{S_t} = \mu dt + \sigma dB_t, \quad \text{where } dS_t = S_{t+\Delta t} - S_t$$

- It can be shown that $\mu dt + \sigma dB_t \sim \text{Normal}(\mu \cdot \Delta t, \sigma^2 \cdot \Delta t)$

- This notation is only symbolic as dB_t does not “exist”:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t \quad \text{or} \quad dS_t = \mu S_t dt + \sigma S_t dB_t$$

- The above can be legitimately viewed as the following integral equation:

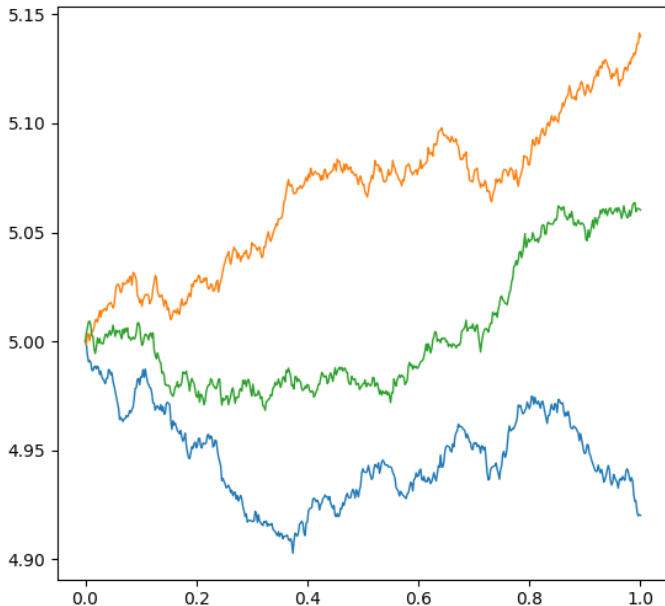
$$S_t - S_0 = \int_0^t \mu S_k dk + \int_0^t \sigma S_k dB_k.$$

- Notice that we have came up with a random variable that models the stock price:

$$S_t(\omega) = S_0 + \int_0^t \mu S_k(\omega) dk + \int_0^t \sigma S_k(\omega) dB_k(\omega).$$

- The stochastic processes S_t that solves the above equation is called a *Geometric Brownian Motion*.

Geometric Brownian Motion: $dX_t = \sigma X_t dW_t + \mu X_t dt$
(a, b) = (0, 1), $X_a = 5$, $\sigma = 0.01$, and $\mu = 0.001$
we subdivide the domain $n = 500$ times



Itô's Integral (a special case – *Wiener integral*)

Suppose that f is a nice continuous function. Then we can define:

$$\int_0^t f(s)dB_s(\omega) := \lim_{\|P\| \rightarrow 0} \sum_{i \in P} f(t_{i-1})(B_{t_i}(\omega) - B_{t_{i-1}}(\omega)),$$

where $P = t_0 = 0, t_1, t_2, \dots, t_n = t$ is a partition of $[0, t]$. In other words, the Itô's integral in this case the left point Riemann sum.

In general, this formula is not true because the increments

$$dB_{t_i}(\omega) := B_{t_i}(\omega) - B_{t_{i-1}}(\omega)$$

are very bad.

Stochastic Heat Equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \frac{\partial^2}{\partial x^2} u(t, x) = \lambda u(t, x) \frac{\partial^2}{\partial t \partial x} W(t, x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R} \end{cases}$$

The solution to the stochastic heat equation is given through the following integral equation:

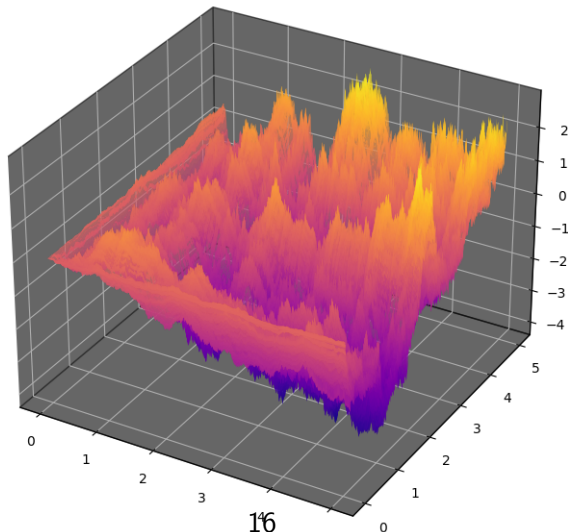
$$u(t, x) = \int_{\mathbb{R}} G(t, x - y) f(y) dy + \lambda \int_0^t \int_{\mathbb{R}} G(t - s, x - y) u(s, y) W(ds, dy),$$

where

$$G(t, x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{|x|^2}{2t}\right)$$

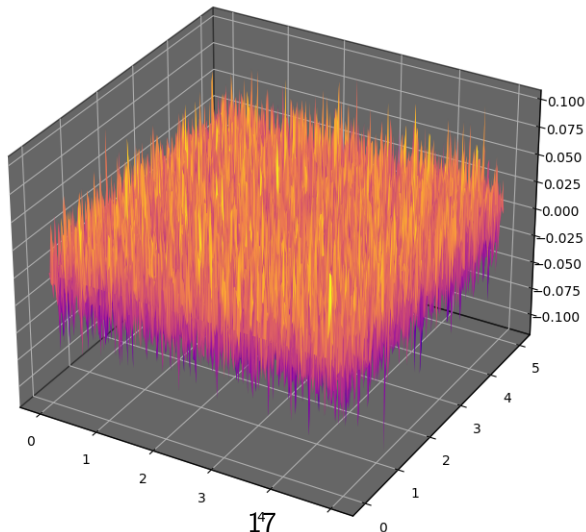
Sample Path of a Brownian Sheet, $W(t, x)$

Brownian Sheet



Approximation of a Space-Time White Noise Differential

Space-time white noise: $\dot{W}(t, x) \approx \int_{t-\delta t}^t \int_{x-\delta x}^x dW(t, x)$



Random vs Deterministic

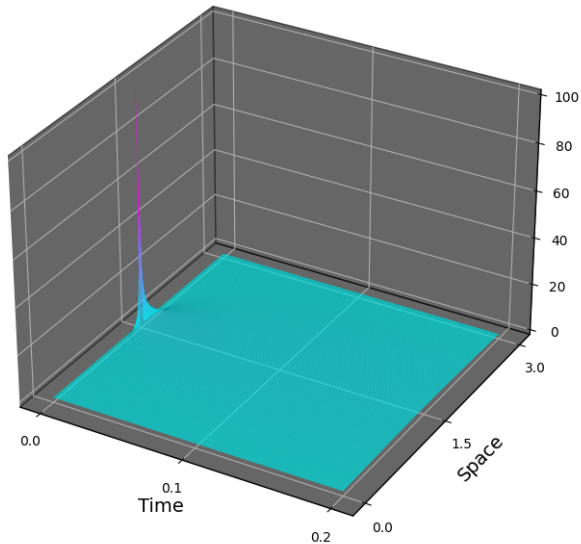
- ▶ The solution to the stochastic heat equation is a stochastic process.

$$u(t, x)(\omega_1), \quad u(t, x)(\omega_2), \quad \cdots, \quad u(t, x)(\omega_n).$$

- ▶ The random solution only exists implicitly through the integral equation.
- ▶ $\frac{\partial}{\partial x} u(t, x)$ and $\frac{\partial}{\partial t} u(t, x)$ does not exists in the random case with space-time white noise.
- ▶ Interesting 'blow up' happens in the random case. This actually has real-world application.

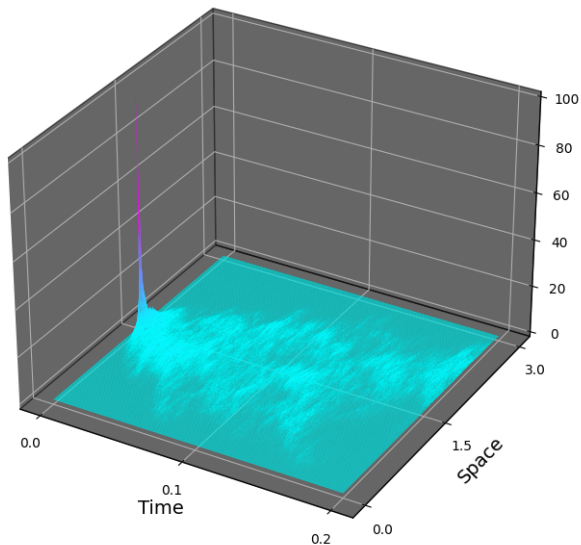
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 0.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



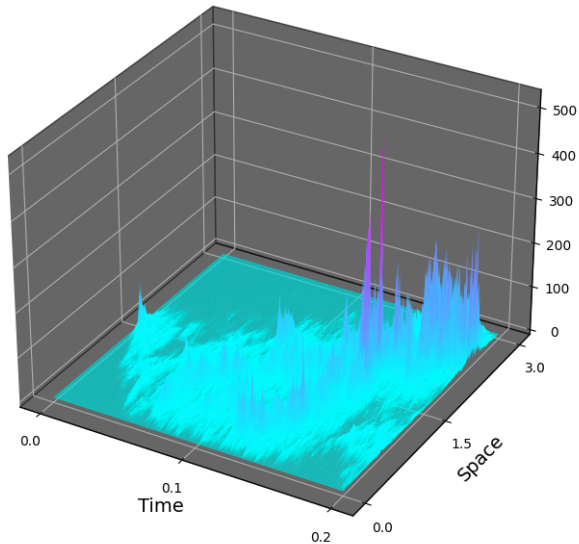
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 1.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



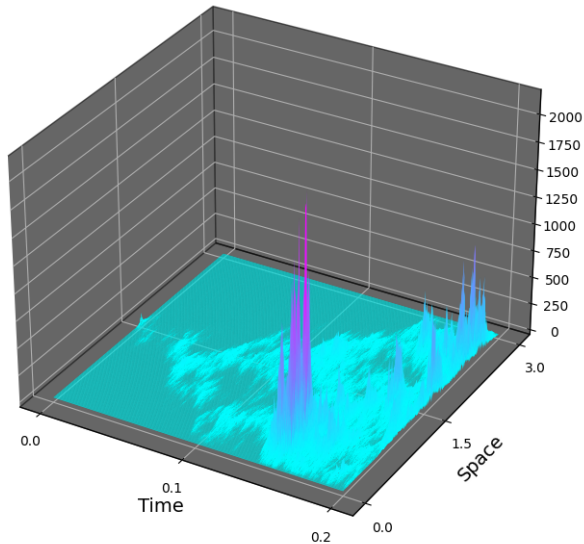
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.05u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.15u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



Growth of the paths and moments

- Can we accurately describe the creation of these large peaks? In other words, how can we describe the sample paths of the solution:

$$t \mapsto u(t, x)(\omega) \quad \text{and} \quad x \mapsto u(t, x)(\omega)$$

- This is a hard question, a current open problem is calculate the following limit

$$\lim_{t \rightarrow \infty} \frac{\log |u(t, x)|}{t^K}.$$

- This limit has been calculated in the following sense for the SHE (and also other SPDEs which we will see below)

$$\lim_{t \rightarrow \infty} \frac{\log [\mathbb{E}(|u(t, x)|^p)]}{t^K}, \quad p \geq 2.$$

Growth of moments for the stochastic heat equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \frac{1}{2} \frac{\partial^2}{\partial x^2} u(t, x) = \sqrt{\lambda} u(t, x) \frac{\partial}{\partial x} W(x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = 1 & x \in \mathbb{R} \end{cases}$$

Theorem (Hu 2002)

For the above equation, we can prove that

$$\lim_{t \rightarrow \infty} \frac{\log [\mathbb{E}(|u(t, x)|^p)]}{t^3} = \frac{p(p-1)\lambda^2}{24}$$

which means that

$$t \mapsto \mathbb{E}(|u(t, x)|^p) \asymp \exp\left(\frac{p(p-1)\lambda^2}{24} t^3\right), \quad \text{as } t \rightarrow \infty.$$

So the moments grow exponentially in time.

The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sqrt{\theta} u(t, x) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (\text{ISHWE})$$

- ∂_t^b is the *Caputo* fractional derivative
- $(-\Delta)^{a/2}$ is the fractional Laplacian of order $a \in (0, 2]$
- I_t^r is the Riemann-Liouville fractional integral of order $r \geq 0$
- $\dot{W}$ is a time-independent Gaussian noise defined on $\phi, \psi \in C_0^\infty(\mathbb{R}^d)$:

$$J(\phi, \psi) := \mathbb{E}(W(\phi)W(\psi)) = \int_{\mathbb{R}^d} dx \, \phi(x)|x - y|^{-\alpha}\psi(y)$$

Fractional Differential Operators

- We recall a property of the standard Laplacian, $\Delta = \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$:

$$(-\Delta)f(x) = \left((|\cdot|^2)\hat{f}(\cdot) \right)^\vee(x).$$

This gives motivation for the fractional Laplacian:

$$(-\Delta)^{\alpha/2}f(x) := \left((|\cdot|^\alpha)\hat{f}(\cdot) \right)^\vee(x),$$

- The Caputo derivative is defined as

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^{(m)}(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

Mild Solution

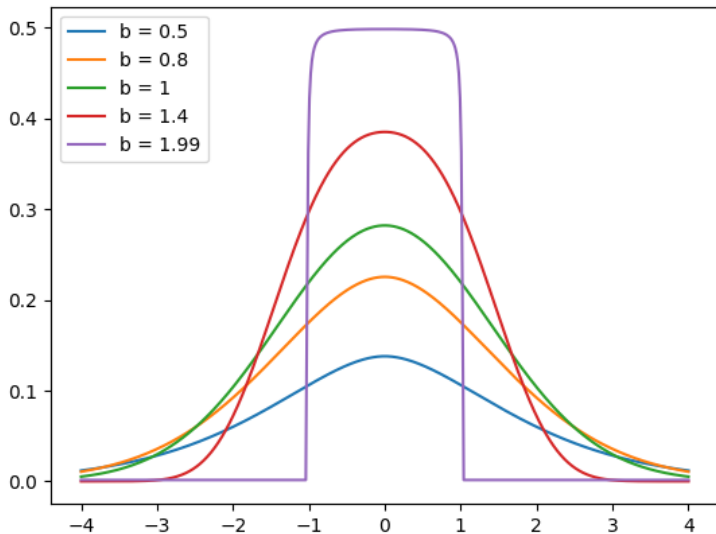
- The solution to the ISHWE is any processes that satisfies the following stochastic integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (0.1)$$

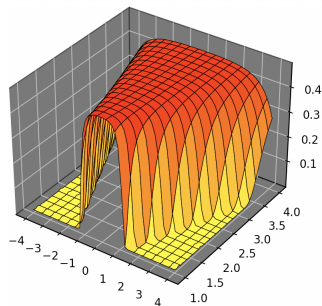
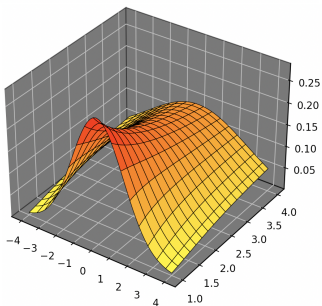
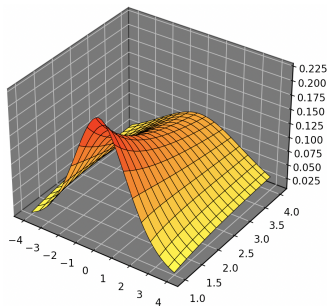
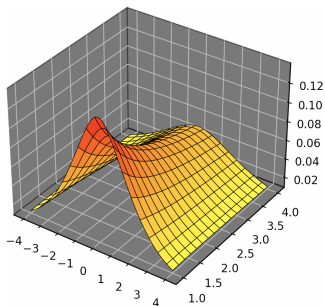
- The kernel G is defined through the Fox-H function:

$$\begin{aligned} & G_{a,b,r,\nu,d}(t, x) \\ &= \pi^{-d/2} |x|^{-d} t^{b+r-1} H_{2,3}^{2,1} \left(\frac{|x|^a}{2^{a-1} \nu t^b} \left| \begin{array}{l} (1, 1), (b+r, b) \\ (d/2, a/2), (1, 1), (1, a/2) \end{array} \right. \right). \end{aligned}$$

$G(1, x)$ with $a = 2, \gamma = 0, \nu = 2, d = 1$ and $.5 \leq b < 2$



$G(t, x)$ with $b = .5, .8, 1, 1.9$ and $1 \leq t \leq 4$



Global Solution

$$(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right]$$

Definition

$u(t, x)$ is a *global solution* to the (ISHWE) if $\|u(t, x)\|_p < \infty$ for any $0 < t < T$ and $x \in \mathbb{R}^d$ where $T > 0$ is arbitrary.

Theorem (Chen-E. '21+)

A global solution exists provided G is nonnegative and

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

Local Solution

Definition

$u(t, x)$ is a *local solution* to the (ISHWE) if there exists $0 < T_a \leq T_b < \infty$ such that $\|u(t, x)\|_2 < \infty$ when $0 < t < T_a$ and $\|u(t, x)\|_2$ D.N.E. (does not exist) for $t > T_b$.

Theorem (Chen-E. '21+)

A local solution exists provided G is nonnegative and if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r) - 1] \leq d.$$

In this case, a unique $L^p(\Omega)$ solution exists for $t \in (0, T_p)$ where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}}$$

and the solution does not exist for $t > T_2$.

Special Form of the Solution

- The time independent noise and the constant initial conditions allows to prove the following explicit representation of the solution:

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$$

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

- Moreover, we can show that the convergence of the above series is equivalent to showing the following:

$$\|u(t, x)\|_2^2 = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 < \infty$$

Exact Moment Lyapunov Exponent

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $p \geq 2$ fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $t \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_1 t^\beta)$, in other words, the moments will grow exponentially in time, and we obtain the exact expression for the constant K_1 .

The Case of a time-dependent Noise and super-linear diffusion term

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sigma(u(t, x)) \dot{W}(t, x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (0.2)$$

- The diffusion coefficient has super-linear growth:

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta \quad \text{as } |x| \rightarrow \infty.$$

- The noise is white in time and either white or colored in space, ie for $\phi, \psi \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$

$$\mathbb{E}(W(\phi)W(\psi)) = \begin{cases} \int_0^\infty ds \int_{\mathbb{R}^d} dx \phi(s, x) \psi(s, x) \\ \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) |x - y|^{-\alpha} \psi(s, y). \end{cases}$$

Results when $|\sigma(x)| \leq |\sigma(0)| + L(\sigma)|x|$

- A global solution exists when $\rho := 2b + 2r - 1 - bd/a > 0$ and $d < 2a$.
- When a global solution exists, then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

Then for any $T > 0$,

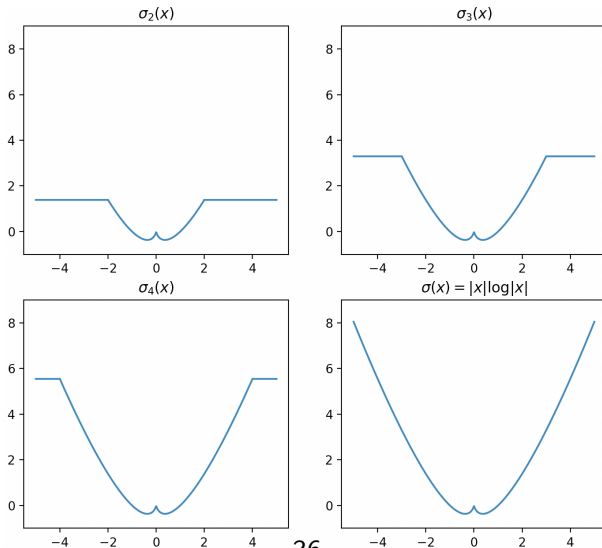
$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)}\right]^p, \quad t \in [0, T]. \quad (0.3)$$

- The solution $u(t, x)$ is continuous in t and in x .

Existence of the Solution for Super-linear σ Term

We consider a truncated version of the equation where σ replaced by σ_N :

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x|\leq N\}} + \sigma(N)\mathbf{1}_{\{x>N\}} + \sigma(-N)\mathbf{1}_{\{x\leq -N\}}.$$



Sketch

- We see that σ_N is Lipschitz:

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|,$$

so there exists a unique solution which is Hölder continuous, which we denote as $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$.

- When

$$2b + 2r - 1 - bd/a > 0 \quad \text{and} \quad d < 2a,$$

$u_N \rightarrow u$ which satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, \text{ a.s.}$$

- The convergence is not $\mathbb{E}(|u_N - u|^2) \rightarrow 0$, which is the usual convergence that is proven. What we have is

$$\lim_{N \rightarrow \infty} \mathbb{P}(|u_N - u|) \rightarrow 0.$$

- So far not much is known about u other than existence and uniqueness.